Data Selection and Training Budget for Reasoning Supervised Fine-Tuning of Qwen2.5-3B-Instruct

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Abstract

We study training-data selection for supervised fine-tuning (SFT) of Qwen2.5-3B-Instruct on long chain-of-thought (CoT) demonstrations. From a 100K subset of AceReason-1.1-SFT, we select 15K examples by domain filtering, exact deduplication, length filtering to the training context, and stratified sampling over source and length, and compare against a uniformly random 15K subset. On eight math and science benchmarks under the official evaluation protocol, our final model, trained for five epochs on the selected set, reaches a macro pass@8 of 0.590 and pass@1 of 0.269 over the six leaderboard benchmarks, versus 0.539 and 0.262 for the base model and 0.549 and 0.246 for the random baseline. The 95% confidence intervals of a paired bootstrap exclude zero for the pass@8 gain over the base model and for the pass@1 gain over the random baseline. The improvement comes mainly from fewer generations truncated at the 32,768-token limit.

1 Introduction

The project [reds-lab, 2026] fine-tunes Qwen2.5-3B-Instruct [Yang et al., 2024] on 15,000 examples drawn from a 100K subset of AceReason-1.1-SFT [Liu et al., 2025], whose responses are long CoT traces generated by DeepSeek-R1, and asks whether a principled selection of the 15K examples outperforms random selection. Full-parameter SFT on such traces teaches the student to imitate step-by-step reasoning, self-verification, and the boxed-answer format, but a 3B model reproduces long traces imperfectly [Li et al., 2025].

We construct a selected training set with a label-free pipeline that needs no model inference (Section 2), train on it with the same recipe as the random baseline, and continue training for two additional epochs. The final model improves macro pass@8 by 0.051 over the base model and pass@1 by 0.023 over the random baseline, and we trace both gains to a lower truncation rate of the generated reasoning (Section 4).

2 Data selection strategy

The training pool contains 100,000 examples, of which 67,432 are math and 32,568 are code. The random baseline samples 15,000 examples uniformly with a fixed seed and no filtering; 556 of them exceed the 16,384-token training context and are truncated, so their final answers are removed from the training target.

Our selected set applies four steps in order. First, *domain filtering* keeps only math examples, since all target benchmarks are math or science and code traces are off-target. Second, *exact deduplication* removes repeated prompts (7,730 among the math examples) and keeps the first occurrence, which increases prompt diversity [Liu et al., 2025]. Third, *length filtering* keeps only examples whose full conversation fits the 16,384-token context under the Qwen chat template (removing 2,292), so that

Table 1: Training hyperparameters of the final model and the random baseline.

	Final model	**Random baseline**
Training data	Selected 15K (107.6M tokens/epoch)	Random 15K (113.4M tokens/epoch)
Epochs (steps)	5 (9,375)	3 (5,625)
Learning rate	5×10^{-6}, cosine, no warmup; epochs 4–5 decay from 1.7×10^{-6} to 0	5×10^{-6}, cosine, no warmup
Tokens seen	538.2M	338.8M
Hardware (time)	4×L40S (8.6 h) + 4×A100 (4.4 h)	4×L40S (9.2 h)
Base model	Qwen2.5-3B-Instruct, full-parameter SFT, loss on response tokens	
Optimizer	AdamW, $\beta = (0.9, 0.999)$, weight decay 0, gradient clipping 1.0	
Global batch size	8 (4 GPUs × micro-batch 1 × gradient accumulation 2)	
Max. sequence length	16,384 tokens, Qwen chat template	
Precision, kernels	bf16, DeepSpeed ZeRO-3, FlashAttention-2, Liger kernel	

every training target ends with its final answer and end-of-turn token. Fourth, *stratified sampling* with proportional allocation draws 15,000 of the 57,410 remaining examples over strata defined by data source (NuminaMath-CoT, OpenMathReasoning) and total-length band (up to 4K, 4–8K, and 8–16K tokens), which preserves the source mix and the difficulty spectrum implied by solution length. The selected set has 107.6M training tokens per epoch (113.4M for random), a median of 6.5K tokens per example, and 2,252 examples in common with the random set.

The design follows evidence that a small, curated fine-tuning set can match or exceed a larger unfiltered one for alignment [Zhou et al., 2023] and reasoning [Muennighoff et al., 2025, Ye et al., 2025], and the question filtering and source mixing studied for reasoning data recipes [Guha et al., 2026]. A data-contamination check found six pool prompts that match a benchmark question verbatim; one falls into each 15K set and is kept and disclosed rather than removed from only one arm.

3 Experimental setup

Training. Both models are trained by full-parameter SFT with LLaMA-Factory [Zheng et al., 2024], using DeepSpeed ZeRO-3 [Rajbhandari et al., 2020], FlashAttention-2 [Dao, 2024], and the Liger fused linear cross-entropy kernel [Hsu et al., 2024], with the loss computed on response tokens only. Table 1 lists the hyperparameters. The final model is trained for three epochs and then resumed from the epoch-3 weights, optimizer, and scheduler states for epochs 4 and 5 on a five-epoch cosine schedule.

Evaluation. We evaluate on AIME24, MATH-500, CN-Math 2024, Kaoyan, AMC, Minerva, OlympiadBench, and GPQA with the course harness, which generates with vLLM [Kwon et al., 2023] and grades the last boxed answer by symbolic equivalence. Each question receives $n = 8$ samples at temperature 0.6, top-p 0.95, and a maximum of 32,768 generated tokens. pass@8 counts a question as solved if any of the eight samples is correct, the pass@k metric of Chen et al. [2021] with $k = n$, and pass@1 is the mean per-sample accuracy. The macro score averages the six leaderboard benchmarks available to us (CN-Math 2024, Kaoyan, AMC, Minerva, OlympiadBench, and GPQA) with equal weight; AIME24 and MATH-500 are reported but excluded from the leaderboard average. Differences between models are reported with 95% confidence intervals from a paired bootstrap with $B = 2000$ that resamples questions and keeps the eight samples of a question together [Efron and Tibshirani, 1993, Miller, 2024].

4 Results and analysis

Table 2 reports all eight benchmarks. Fine-tuning on the random subset changes the macro pass@8 of the base model by only $+0.011$ $[-0.042, +0.062]$ and lowers its pass@1 by 0.016 $[-0.036, +0.004]$. The final model improves macro pass@8 over the base model by $+0.051$ $[+0.009, +0.095]$, the only gain over the base model whose interval excludes zero, and over the random baseline by $+0.040$ $[-0.004, +0.086]$ in pass@8 and $+0.023$ $[+0.009, +0.038]$ in pass@1. Per benchmark, the largest

Table 2: pass@8 and pass@1 on the eight benchmarks. N: number of questions; *: leaderboard benchmark included in the macro average.

	Base		Random 15K		Final model	
Benchmark (N)	pass@8	pass@1	pass@8	pass@1	pass@8	pass@1
AIME24 (30)	0.167	0.071	0.333	0.071	0.233	0.075
MATH-500 (500)	0.840	0.653	0.884	0.647	0.878	0.659
CN-Math 2024* (30)	0.367	0.104	0.333	0.079	0.333	0.104
Kaoyan* (199)	0.548	0.219	0.653	0.292	0.678	0.298
AMC* (40)	0.650	0.391	0.575	0.344	0.775	0.381
Minerva* (272)	0.452	0.287	0.471	0.255	0.456	0.273
OlympiadBench* (675)	0.498	0.277	0.526	0.278	0.538	0.301
GPQA* (198)	0.717	0.294	0.737	0.230	0.758	0.258
Macro (*)	0.539	0.262	0.549	0.246	**0.590**	**0.269**

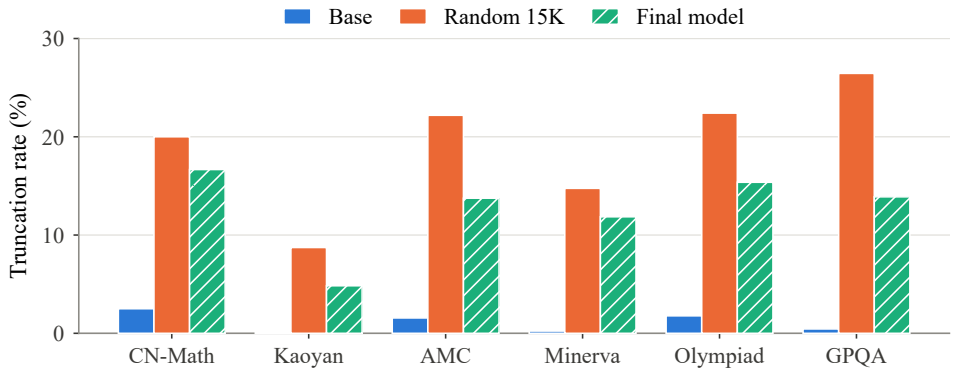


Figure 1: Truncation rate, the share of samples that reach the 32,768-token generation limit, on the six leaderboard benchmarks.

pass@8 gains are on Kaoyan ($+0.131$ $[+0.070, +0.191]$ over the base model) and AMC ($+0.200$ $[+0.050, +0.375]$ over the random baseline), and the clearest pass@1 gain over the random baseline is on OlympiadBench ($+0.023$ $[+0.012, +0.035]$). On AIME24, which is not in the macro average, the final model is below the random baseline (-0.100 $[-0.200, 0.000]$).

Truncation is the main source of error. Both fine-tuned models generate CoT traces 8–16 times longer than the base model, consistent with the overthinking of long-reasoning models [Chen et al., 2024]. As shown in Figure 1, 8.7–26.5% of the random baseline's samples on the leaderboard benchmarks reach the 32,768-token limit before giving an answer, and a truncated sample is almost always graded wrong (accuracy 0–4%). The final model lowers the truncation rate on every leaderboard benchmark, to 4.8–16.7%, while the accuracy of completed samples stays comparable (e.g., 0.355 versus 0.352 on OlympiadBench and 0.309 versus 0.296 on Minerva). Fewer truncations therefore translate directly into higher pass@1 and pass@8. Two factors are likely to contribute: length filtering ensures that every training target is a complete solution with an answer, whereas the random set contains 556 truncated targets, and longer training continues to reduce run-on generation. On a held-out set of 200 pool questions in neither training set, the truncation rate of the selected run falls from 55% after epoch 1 to 24% after epoch 3 and 16% after epoch 5, in line with the finding that SFT accuracy on this data family improves up to about five epochs [Liu et al., 2025].

Data selection versus training budget. The final model is trained for five epochs and the random baseline for three, so their difference combines data selection and training budget. At an equal budget of three epochs, the selected set reaches 0.566 and 0.244, a pass@8 gain of $+0.016$ $[-0.024, +0.056]$ over the random baseline that lies within question-sampling noise. The gain of the final model is therefore best attributed to the selected data and the longer training together.

Limitations. Each configuration is trained once, so the intervals cover question sampling but not training-seed variance. The five-epoch schedule was chosen after the three-epoch results were known, and the final model is selected on the benchmarks it is reported on; AIME25, which is on the course leaderboard but not available to us, is the only unseen test. All training data come from the provided pool. Adding external data is allowed but would require decontamination; short, curated reasoning sets [Muennighoff et al., 2025, Ye et al., 2025], which suit small students [Li et al., 2025], and science data for GPQA are natural extensions. Additional selection methods we evaluated, including a learning-progress selector, are reported in the full report in the repository.

5 Conclusion

A label-free selection of complete, deduplicated math demonstrations, combined with five epochs of SFT, raises the macro pass@8 of Qwen2.5-3B-Instruct from 0.539 to 0.590 and its pass@1 from 0.262 to 0.269, and outperforms the random baseline (0.549 and 0.246), mainly by reducing truncated generations. Code, configurations, and evaluation logs are available at `https://github.com/yuan-zhou-git/Course_Project_Trustworthy_ML`; the final model and datasets are available at `https://huggingface.co/yuan-zhou-hf`.

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